HDOC DAY 4

QUESTION -1 The following problem is to find the product of two integers without using the multiplication operator *. Your

teacher has given an incorrect algorithm and an incorrect flowchart for this problem. Study both carefully and complete the tasks given below.

SOLUTION 1:

1. Mistakes in the given algorithm

The common mistakes in an incorrect algorithm for this problem are:

1. It may assume both numbers are positive and therefore fail for negative numbers.
2. It may not handle zero correctly.
3. It may use repeated addition of the first number without considering which number is smaller, resulting in unnecessary additions.
4. It may not correctly determine the sign of the final answer.
5. It may perform addition using a negative value in a way that gives incorrect results.
6. It may not initialize product to 0.
7. The loop condition may be incorrect, causing too many or too few additions.

2. Corrected Algorithm

Suppose the two integers are A and B.

Algorithm: Product of Two Integers Without *

1. Start.
2. Input two integers A and B.
3. If A = 0 or B = 0:
   - Set Product = 0.
   - Display Product.
   - Stop.
4. Find the absolute values:
   - X = |A|
   - Y = |B|

5. If X < Y:
    - Count = X
    - Number = Y
6. Otherwise:
    - Count = Y
    - Number = X
7. Set:
    - Product = 0
    - i = 0
8. Repeat while i < Count:
    - Product = Product + Number
    - i = i + 1
9. Check the signs of the original numbers:
    - If one number is negative and the other is positive, make Product negative.
    - Otherwise, keep Product positive.
10. Display Product.
11. Stop.

3. Why This Algorithm Works

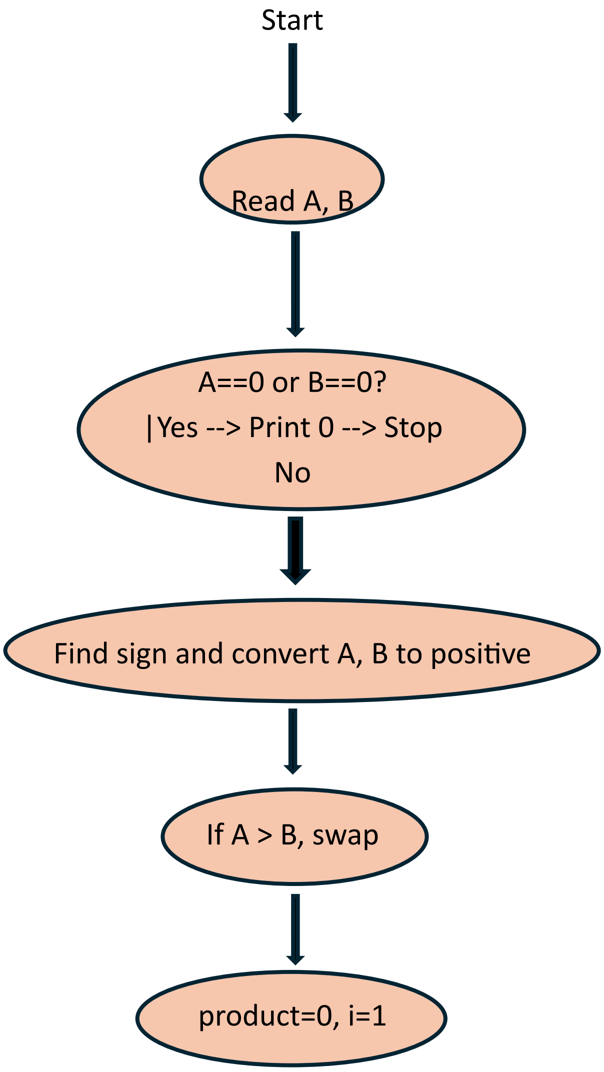
A	**B**	**PRODUCT**
+ve	+ve	+ve
+ve	-ve	-ve
-ve		-ve
-ve	-ve	+ve
Any number	zero	zero

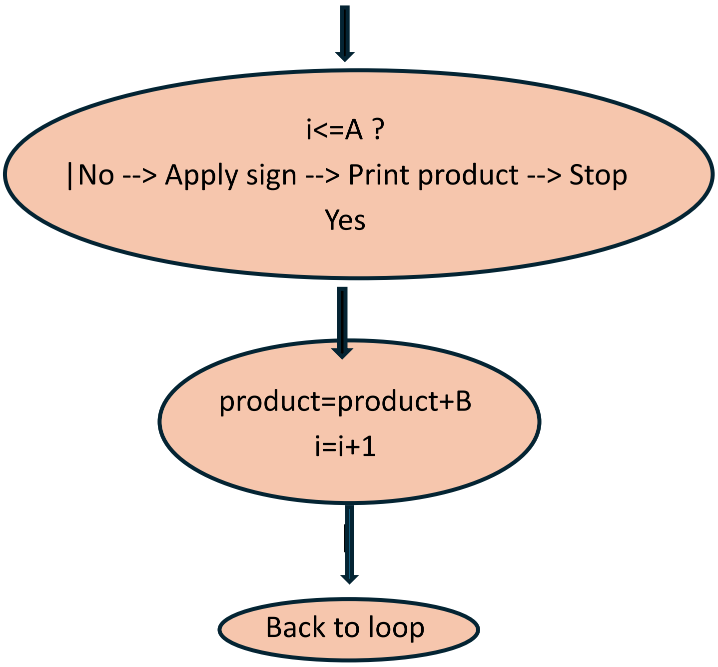
4. Mistakes in the Given Flowchart

The incorrect flowchart may contain the following mistakes:

1. It may not check whether either number is zero.
2. It may not handle negative numbers.
3. It may not convert negative values into absolute values before repeated addition.
4. It may not check which absolute value is smaller.
5. It may therefore perform unnecessary additions.
6. It may not correctly determine the sign of the final product.
7. The loop condition or increment may be incorrect.
8. The flow of arrows may not return correctly to the loop condition.

5. Corrected Flowchart





6. Verification Using Test Cases

INPUT	EXPECTED	MINIMUM ADDITIONS	VERIFIED
12,3	36	3	YES
3,12	36	3	YES
-6,5	-30	5	YES
6,-5	-30	5	YES
-9,-4	36	4	YES
-4,-9	36	4	YES
0,8	0	0	YES
8,0	0	0	YES
0,-7	0	0	YES
-7,0	0	0	YES
0,0	0	0	YES
1,-7	-7	1	YES
-1,-9	9	1	YES

Hence,all the the evaluation and verification has been given above.